



OMNIX QUANTUM LTD

OMNIX-RTE-001

Runtime Treasury Execution Trace
ADR-201 . RFC-ATF-1 through RFC-ATF-6

SCENARIO

Autonomous Treasury Approval -- Cross-border Liquidity Release

USD 50,000,000 - SWIFT MT202 / XRPL RLUSD

Package: OMNIX-RTE-001-77B1453F819D4F52

DUAL-PATH RESULT SUMMARY

PATH DANGEROUS

Result	HALT -> OSG REJECTED
Budget	42% (drift -58%)
CES	42.0 CRITICAL
MAS	0.24 HALT
Settlement	BLOCKED

PATH ADMISSIBLE

Result	ADMITTED -> OSG APPROVED
Budget	88% (recertified)
CES	95.5 NOMINAL
MAS	0.94 ALIGNED
Settlement	RELEASED USD 50,000,000

Generated	2026-05-30 10:38:59 UTC
PQC Algorithm	ML-DSA-65 (Dilithium-3, FIPS 204)
Invariants	30 demonstrated
Standard	RFC-ATF-1 through RFC-ATF-6 - ADR-201 - FIPS 204
PQC Public Key SHA-256	E4D63A7F20A43ECF582C5BD94B2C9D7081E454E9ACB68349 . . .

§2 - Scenario Overview

Treasury Request

Request ID	REQ-76BECDF8B042
Amount	USD 50,000,000
Settlement Rail	SWIFT MT202 / XRPL RLUSD
Counterparty	-
Counterparty BIC	-
Account (IBAN)	-
Purpose	-
FX Rate	0 EUR/USD
FX Band	-
Risk Level	-
Regulatory Epoch	-

Source State Integrity

source_state_hash4c6bab9879912977a0b514f5f42b9f5b43d1cd32bca22baee5376fa6966922be

Policy Constraints (Governing Mandate)

Max Single-Tx Amount	USD 50,000,000 (requested: USD 50M)
Max FX Deviation	±0%
Counterparty Whitelist	Required - EUROBANK-COUNTERPARTY-001 on list
Dual Approval	Required for amounts > USD 5M
Settlement Window	-

Compliance Tags

Governance Session Parameters

Agent ID	TREASURY-AGENT-OMNIX-001
Agent Role	AUTONOMOUS_TREASURY_EXECUTOR
ATF Layer	L1 (Identity) + L2 (Delegation) + L3 (Temporal) + L4 (Continuity) + BEV L6
Session Type	DUAL-PATH - Dangerous + Admissible
PQC	Dilithium-3 (ML-DSA-65, FIPS 204) . Ephemeral keypair per package
Standards	RFC-ATF-1 -> RFC-ATF-6 . ADR-201

§3 - Governance Architecture

The OMNIX-RTE-001 package exercises the complete ATF + BEV governance stack across two parallel execution paths. Each path traverses an identical 8-step chain, producing a PQC-signed artifact at every transition. The divergence occurs at step 3 (Runtime Evaluation) when the Continuity Execution Score (CES) and MIVP Mandate Alignment Score (MAS) determine whether the session proceeds to settlement or is halted.

8-Step RTE-001 Chain

1_SOURCE_STATE	Source State capture + Temporal Context Snapshot (TCS)
2_AUTHORITY	Delegation Receipt (DR) + Mandate Binding Record (MBR)
3_RUNTIME	Continuity Execution Score (CES) + MIVP Mandate Alignment Score (MAS)
4_COUNTERFACTUAL	Counterfactual Fork Records (CFRs) + Counterfactual Attestation Token (CAT)
5_VERDICT	Temporal Admissibility Record (TAR) - binding verdict
6_GATE	OSG Settlement Gate - ValidationReceipt (VR) + PoGC (admissible only)
7_EXECUTION	Refusal Receipt (dangerous) / BAR + CTCHC + Outcome (admissible)
8_POST_EXECUTION	Replay Proof + Post-execution TCS - terminal forensic seal

Artifact Inventory

Artifact	Name	Hash / Sig Profile	Invariant
DR	Delegation Receipt	SHA-256 + compact + sign(content_hash)	ATF-INV-001
MBR	Mandate Binding Record	SHA3-256 + default + compact sig	MIVP-INV-001
MAS	Mandate Alignment Score	SHA3-256 + default + compact sig	MIVP-INV-003
RCR	Runtime Continuity Record	SHA3-256 + default + compact sig	ATF-INV-003
CFRx5	Counterfactual Fork Records	SHA3-256 per CFR + fragility	CGE-INV-002
CAT	Counterfactual Attestation Token	SHA3-256 + cfr_root_hash + compact sig	CGE-INV-007
TAR	Temporal Admissibility Record	SHA-256 + compact + sign(content_hash)	ATF-INV-004
OSG-VR	OSG Validation Receipt	SHA3-256 + default + compact sig	OSG-INV-001
BAR	Behavioral Anchor Record	SHA3-256 6-field canonical + default sig	BEV-INV-002
CTCHC	Cross-Turn Coherence Hash Chain	SHA3-256 per link + 7-field seal hash	BEV-INV-013
PoGC	Proof of Governance Certificate	SHA3-256 + compact sig	PoGR-INV-003
MBR Seal	MBR Session Seal	SHA3-256 + compact sig + tier	MIVP-INV-007
TCS	Temporal Context Snapshot	SHA3-256 + default + compact sig	TGB-INV-001
Replay	Post-execution Replay Proof	SHA3-256 + default + compact sig	RTE-INV-007

§4 - Dual-Path Comparison		
Dimens	PATH DANGEROUS	PATH ADMISSIBLE
Authori	42% (drift ?58%)	88% (recertified)
DR Expi	~22 min (near-expiry)	4h (fresh)
CES Sc	42.05	95.50
CES Ba	CRITICAL	NOMINAL
MIVP M	0.24 (HALT < 0.30)	0.94 (ALIGNED > 0.65)
MIVP Ve	HALT	ALIGNED
TAR St	ADMITTED	-
OSG Ve	REJECTED	APPROVED
BAR St	VALID	VALID
CTCHC	True	True
PoGC Is	No (HALT path)	POGC-FB915D9A553A49F3
Mandati	UNCERTIFIED	MANDATE-BOUND
Settlem	BLOCKED	RELEASED - T+2 clearing
Replay	HALTED	CLOSED

§5 - Authority Layer: DR + MBR (Both Paths)

PATH DANGEROUS - Authority Layer

Delegation Receipt (DR) . ATF-INV-001

Delegation ID	ATFDR-BDA2ECB324EA45B0
Budget Granted	42.0%
Budget Delegator	100.0%
MAR Reduction	?58.0%
Issued At	-
Expires At	2026-05-30 11:00:35 UTC
PQC Algorithm	Dilithium-3 (ML-DSA-65)
content_hash	b0db7888f52215a23c4493f071f057d631f107e3902215fe613b35c6a678bbcc
pqc_signature	bhunJRF4gsbQEkmSYZW4VpF31+Q1DF5UzSSVChG0Adwk/uxbzwH2d/B9qMxYcn866RpRqSwT

Mandate Binding Record (MBR) . MIVP-INV-001

MBR ID	MBR-23A494C868B54C7B
Session ID	SESSION-B105A1CF98324960
HALT Threshold	0.30
WARN Threshold	0.65
Issued At	2026-05-30 10:38:36 UTC
mbr_content_hash	90a40853d65bc2573bb6b6d64680c3a25283a63e62e6b4661f919054b590ff87

PATH ADMISSIBLE - Authority Layer

Delegation Receipt (DR) . ATF-INV-001

Delegation ID	ATFDR-383F317749984B5F
Budget Granted	88.0%
Budget Delegator	100.0%
MAR Reduction	?12.0%
Issued At	-
Expires At	2026-05-30 14:38:44 UTC
PQC Algorithm	Dilithium-3 (ML-DSA-65)
content_hash	f8ee8c665c87923a839046db8fb5a4a3a8ff786c3971ec4a6e15d6a53ee3ce6d
pqc_signature	JyhZ+qkums7TcB6in8lGG0tJ0Er0KctfRFqZ0maIaHcMrfFAJ0bqaXCgZuLMzPz/Zy+TokBs

Mandate Binding Record (MBR) . MIVP-INV-001

MBR ID	MBR-7E2B9E0FAF044CCF
Session ID	SESSION-F2B01C97D00141FC
HALT Threshold	0.30
WARN Threshold	0.65
Issued At	2026-05-30 10:38:45 UTC
mbr_content_hash	8210d16bf60ed88544ddbdaacfb6411b5e312ddc482204fd28020fef959dd84

§6 - Runtime Evaluation: RCR / CES + MAS (Both Paths)

PATH DANGEROUS - Runtime Evaluation

Runtime Continuity Record (RCR) . ATF-INV-003

RCR ID	ATFRCR-A27E7658F3454A2F
CES Score	42.05
CES Band	CRITICAL
T (temporal)	9.2% (weight 0.30) -> 2.75
B (budget)	42.0% (weight 0.30) -> 12.60
D (context)	61.5% (weight 0.20) -> 12.30
I (integrity)	72.0% (weight 0.20) -> 14.40
CES Formula	$Tx0.30 + Bx0.30 + Dx0.20 + Ix0.20 = 42.05$
Issued At	2026-05-30 10:38:36 UTC
rcr_hash	b3dabb6d916a85f27736bdd1bb8a755f3de598a6645918afcebae5e9772e49d3

Mandate Alignment Score (MAS) . MIVP-INV-003

MAS ID	MAS-B2AD8C08BC01438B
Turn Index	0
Score	0.2400
Verdict	HALT
Violations	0
Warnings	0
Proxy Guard	CLEAR
Issued At	2026-05-30 10:38:37 UTC
mas_content_hash	68683a736fb27139e56c70de6311126f04e6c27cbfe99c9ea2c9e64643abf777

PATH ADMISSIBLE - Runtime Evaluation

Runtime Continuity Record (RCR) . ATF-INV-003

RCR ID	ATFRCR-F4E7563A4EF84E9E
CES Score	95.50
CES Band	NOMINAL
T (temporal)	100.0% (weight 0.30) -> 30.00
B (budget)	88.0% (weight 0.30) -> 26.40
D (context)	97.0% (weight 0.20) -> 19.40
I (integrity)	98.5% (weight 0.20) -> 19.70
CES Formula	$Tx0.30 + Bx0.30 + Dx0.20 + Ix0.20 = 95.50$
Issued At	2026-05-30 10:38:45 UTC
rcr_hash	786a825d977da74e81c9bc48824e17c8ce6cca35b6026e4f8956cbAAF69e3a28

Mandate Alignment Score (MAS) . MIVP-INV-003

MAS ID	MAS-9550A0C5B60D432B
Turn Index	0
Score	0.9400
Verdict	ALIGNED
Violations	0
Warnings	0
Proxy Guard	CLEAR
Issued At	2026-05-30 10:38:46 UTC
mas_content_hash	12146b5d829d5bdc31c7352c91073af4b14a7f68bee9a410b2fba8fdcd630bc9

§7 - Counterfactual Analysis: CGE CFRs + CAT (Both Paths)

PATH DANGEROUS - Counterfactual Engine (CGE)

Counterfactual Attestation Token (CAT) . CGE-INV-007

CAT ID	CAT-DFA340D40BD04FBC
CFR Count	5
cfr_root_hash	fdfdb944f874aa9ffb82197bd0864b643af05e52886dd77f11dd8776...
Issued At	2026-05-30 10:38:37 UTC

Counterfactual Fork Records (CFRs) . CGE-INV-002 / CGE-INV-003

CFR ID	Label	Fragility	Selected	Blocked Reason
CFR-CAA8FF6C7FE0476F	-	0.94	YES	ADMITTED
CFR-5F606F3055C64265	-	0.08	no	ADMITTED
CFR-43E3892D718A47AD	-	0.71	no	ADMITTED
CFR-6C108B3B712F4C9C	-	0.85	no	ADMITTED
CFR-0EF70865BB9C4E4A	-	0.88	no	ADMITTED

PATH ADMISSIBLE - Counterfactual Engine (CGE)

Counterfactual Attestation Token (CAT) . CGE-INV-007

CAT ID	CAT-69E9E86F6CC44FF9
CFR Count	5
cfr_root_hash	04d74464a56d9cd996f3d1a28b25c964906839657d31e1be76ffc175...
Issued At	2026-05-30 10:38:46 UTC

Counterfactual Fork Records (CFRs) . CGE-INV-002 / CGE-INV-003

CFR ID	Label	Fragility	Selected	Blocked Reason
CFR-0EC66E852DD74C66	-	0.09	YES	ADMITTED
CFR-03AB71FF3B2743BD	-	0.14	no	ADMITTED
CFR-5621266A673D4FFE	-	0.18	no	ADMITTED
CFR-0F077BC97DBA4980	-	0.11	no	ADMITTED
CFR-6FDA348F19BD453D	-	0.13	no	ADMITTED

§8 - DANGEROUS Path: HALT Evidence Chain

Refusal Receipt . RTE-INV-002

Receipt ID	REFUSAL-B7B36D53075444DF
Type	HARD_REFUSAL
Settlement Status	BLOCKED - OSG will independently reject
Mandate Cert	UNCERTIFIED
Rejection Reasons	5
Issued At	2026-05-30 10:38:38 UTC

Rejection Reasons:

- ? CES=42.05 - CRITICAL band (HALT threshold for CRITICAL-risk transaction > USD 10M)
- ? MIVP MAS=0.24 - below halt_threshold=0.3 (MIVP-INV-003)
- ? MIVP proxy_guard_violation: Speed-over-verification optimisation detected
- ? TAR: Authority budget 42% insufficient for CRITICAL-risk USD 50M transaction
- ? CCS: cumulative drift CRITICAL - 3 consecutive drift signals before Turn 1

OSG Validation Receipt (REJECTED) . OSG-INV-001

VR ID	VR-2AE72C5A64574199
Verdict	REJECTED
Fail Closed	True
Issued At	2026-05-30 10:38:38 UTC
vr_content_hash	c1a62188def74346f4206bc9f75f1f140109a8845ba8d47fa49a2bedba47a6e6

MBR Session Seal (UNCERTIFIED) . MIVP-INV-007

Seal ID	MBRSEAL-FE2AF6F5AA454502
Certification Tier	UNCERTIFIED
Issued At	2026-05-30 10:38:38 UTC
seal_content_hash	bcd0dc481e42c48780f30c383108a6522b4da7ddf72d25098c73a2a6de2d181c

Behavioral Anchor Record (BAR) . BEV-INV-001

BAR ID	BAR-EBE108E7B66A413A
Status	VALID
Halt Reason	-
Turn Index	0
content_hash	48c01c9da7eff833ff1b2daec0f4535e9b8e3491606ccae9efbfab87b8765a96
output_hash	84c44d1bc25b04dd1b40413685f06be683c5d9dcf84e1ca1b114acfe39ed099c

CTCHC - Sealed in HALTED State . BEV-INV-013

Chain ID	CTCHC-73F1DB4273EA4349
Turn Count	1
Is Sealed	True
Sealed At	2026-05-30 10:38:43 UTC
seal_hash	156c0548e51fa0b3535d25cf0a317b2227bc941e69dc7761b6f65b663e7c6710

Structural Invariants

execution_occurred	False
settlement_released	False

§9 - ADMISSIBLE Path: Settlement Evidence Chain

TAR . Temporal Admissibility Record - ADMITTED . ATF-INV-004

TAR ID	-
Admission Status	ADMITTED
Execution ns	0
Issued At	-
content_hash	-

TAR Binding Record

Binding ID	BIND-9CE0582CDC3E
Status	-

Proof of Governance Certificate (PoGC) . PoGR-INV-003

PoGC ID	POGC-FB915D9A553A49F3
Mandate Certification	MANDATE-BOUND
TAR Status	ADMITTED
BAR Status	VALID
CTCHC Turn Count	3
Standard	RFC-ATF-6 (ADR-181/182/183/186) . ADR-194 (MIVP)
Regulatory Tags	EU AI Act Art. 9, MiCA Title VI, DORA Art. 11
Issued At	2026-05-30 10:38:59 UTC
content_hash	7df5fc7900a1ef13520e67f9b0b88c7d96c63f09ecd3a38d05c1f4d0bcd0b8de

MBR Session Seal (MANDATE-BOUND) . MIVP-INV-008

Seal ID	MBRSEAL-CD9FA380884F42F7
Certification Tier	MANDATE-BOUND
Issued At	2026-05-30 10:38:59 UTC
seal_content_hash	46882245a6a5ac217333eec84f95465b675c9ebc9baf0420c8d44250888673a7

OSG Validation Receipt (APPROVED) . OSG-INV-001

VR ID	VR-4299C4D39DF04C9C
Verdict	APPROVED
Fail Closed	True
Issued At	2026-05-30 10:38:59 UTC
vr_content_hash	19583d5289fc4504e0886b65a656900dfc6aa37aadeaf1f5b4018e06425ad9b8

Settlement Reference

Amount (USD)	USD 50,000,000
SWIFT MT202 Ref	MT202-E9C25E5EB60E
XRPL TxID	XRPL-434966837583449F95F2B938
Clearing	-
FX Rate	-
Counterparty	-
IBAN	-

Outcome Receipt . RTE-INV-004

Outcome ID	-
Type	ADMISSION_OUTCOME
Mandate Cert	MANDATE-BOUND
Issued At	2026-05-30 10:38:59 UTC
content_hash	07cee44faf1ce82c61d217727f42b990e1239a2834eb75f709721936b064fcaf

§10 - Post-Execution: TCS + Replay Proofs (Both Paths)

PATH DANGEROUS - Post-Execution Forensic Seal

Temporal Context Snapshot (TCS) . TGB-INV-001

TCS ID	ATFTCS-0567AEB03B594F56
Regulatory Epoch	-
Active Frameworks	
Jurisdiction	-
Captured At	-
tcs_hash	da5bb5e9d10834ed8ab6463cbe8eafc15edf2c2934ba233b722bd3458643581b

Replay Proof . RTE-INV-006 / RTE-INV-007

Proof ID	REPLAY-18531B89379D4495
Terminal Status	HALTED
Offline Verifiable	True
CTCHC Chain ID	CTCHC-73F1DB4273EA4349
CTCHC Seal Hash	156c0548e51fa0b3535d25cf0a317b2227bc941e...
Turn Count	1
Issued At	2026-05-30 10:38:44 UTC
proof_content_hash	51bac0170d41afa095ca83fa91739c3c28a724c088edb5024d84d7a83d91e089

PATH ADMISSIBLE - Post-Execution Forensic Seal

Temporal Context Snapshot (TCS) . TGB-INV-001

TCS ID	ATFTCS-1ED4B3A8C05C4A05
Regulatory Epoch	-
Active Frameworks	
Jurisdiction	-
Captured At	-
tcs_hash	2e823ab91bd3354f2716ce9aefcdbbf128196daaf604e4e0a41a24b3da3b048c

Replay Proof . RTE-INV-006 / RTE-INV-007

Proof ID	REPLAY-7F119BD978B24F6E
Terminal Status	CLOSED
Offline Verifiable	True
CTCHC Chain ID	CTCHC-99E4E73D349F406A
CTCHC Seal Hash	d1bf314e48a4b0503fc54623de80c2f88f5c48d5...
Turn Count	3
Issued At	2026-05-30 10:38:59 UTC
proof_content_hash	3a5c2dc25b6ec3e96200587cd6c2ddfbb9e329fa0df85a8cc4fbc8ad45a0efa9e

§11 - Offline Verification: CLI Report

The following verification was produced by the standalone CLI verifier (scripts/verify_treasury_execution_trace.py) with no runtime dependencies. Verification requires only the public key embedded in the package.

Verification Commands

```
python scripts/verify_treasury_execution_trace.py OMNIX-RTE-001_20260530_103859.json
python scripts/verify_treasury_execution_trace.py OMNIX-RTE-001_20260530_103859.json --verify-authority
python scripts/verify_treasury_execution_trace.py OMNIX-RTE-001_20260530_103859.json --verify-continuity
python scripts/verify_treasury_execution_trace.py OMNIX-RTE-001_20260530_103859.json --verify-halt
python scripts/verify_treasury_execution_trace.py OMNIX-RTE-001_20260530_103859.json --verify-settlement
python scripts/verify_treasury_execution_trace.py OMNIX-RTE-001_20260530_103859.json --json
```

Verification Result - 111 / 111 Checks

PASS	[PKG-TYPE] package_type=OMNIX-RTE-001
PASS	[PKG-PATHS] Both dangerous and admissible paths present
PASS	[PKG-PQC] Embedded public_key_b64 present
PASS	[PKG-INV] 30 invariants demonstrated (>=20 required)
PASS	[DR-DAN-HASH] DANGEROUS DR content_hash integrity
PASS	[DR-DAN-SIG] DANGEROUS DR PQC signature (ML-DSA-65)
PASS	[DR-DAN-MAR] DANGEROUS DR MAR: 42.0 <= 100.0 (ATF-INV-001)
PASS	[MBR-DAN-SIG] DANGEROUS MBR PQC signature (MIVP-INV-001)
PASS	[RCR-DAN-CES] CES formula consistent: 42.05
PASS	[RCR-DAN-BAND] CES band CRITICAL (dangerous path)
PASS	[MAS-DAN-SIG] DANGEROUS MAS PQC signature (MIVP-INV-003)
PASS	[CAT-DAN-ROOT] CAT cfr_root_hash covers all CFR IDs
PASS	[CAT-DAN-COUNT] CAT cfr_count >= 3 (got 5)
PASS	[CAT-DAN-SIG] CAT PQC signature (CGE-INV-007)
PASS	[REF-SIG] Refusal receipt PQC signature
PASS	[REF-TYPE] type=HARD_REFUSAL
PASS	[REF-SETTLE] settlement_status=BLOCKED
PASS	[BAR-DAN-HASH] DANGEROUS BAR content_hash (BEV-INV-002)
PASS	[BAR-DAN-SIG] DANGEROUS BAR PQC signature (BEV-INV-004)
PASS	[CHC-DAN-CHAIN] CTCHC link chain continuity (BEV-INV-011)
PASS	[CHC-DAN-SEAL] CTCHC seal_hash covers all links (BEV-INV-013)
PASS	[CHC-DAN-SIG] CTCHC seal PQC signature (BEV-INV-014)
PASS	[OSG-DAN-VERDICT] OSG verdict=REJECTED
PASS	[HALT-EXEC] execution_occurred=False (RTE-INV-002)
PASS	[HALT-SETTLE] settlement_released=False
PASS	[POGC-SIG] PoGC PQC signature (PoGR-INV-003)
PASS	[POGC-BOUND] mandate_certification=MANDATE-BOUND
PASS	[SEAL-ADM-TIER] MBR Seal tier=MANDATE-BOUND (MIVP-INV-008)
PASS	[OSG-ADM-VERDICT] OSG verdict=APPROVED
PASS	[BAR-ADM-SIG] ADMISSIBLE BAR PQC signature (BEV-INV-004)
PASS	[CHC-ADM-SIG] ADMISSIBLE CTCHC seal PQC signature (BEV-INV-014)
PASS	[TAR-ADM-SIG] ADMISSIBLE TAR PQC signature (ML-DSA-65)
PASS	[OUT-SIG] Outcome receipt PQC signature
PASS	[SETTLE-AMOUNT] Settlement amount = USD 50,000,000
PASS	[RPL-DAN-STATUS] Replay proof terminal_status=HALTED
PASS	[RPL-ADM-STATUS] Replay proof terminal_status=CLOSED
PASS	[RPL-DAN-OFFLINE] Replay proof offline_verifiable=True (RTE-INV-007)
PASS	[RPL-ADM-OFFLINE] Replay proof offline_verifiable=True (RTE-INV-007)

VERDICT: 111 / 111 VERIFICATIONS PASS - package integrity confirmed

§12 - Invariants Demonstrated

This package demonstrates 30 governance invariants across RFC-ATF-1 through RFC-ATF-6 and ADR-201. Each invariant is verified by the standalone CLI verifier without platform access.

Invariant
ATF-INV-001: Authority never expands through delegation (MAR enforced at DR issuance)
ATF-INV-002: Every DR PQC-signed by delegator (Dilithium-3)
ATF-INV-005: Receipts immutable once issued (content_hash)
RGC-INV-001: Every RCR anchored to a valid TAR
RGC-INV-002: CES computed from real-time component values
BEV-INV-001: Every governed turn produces a BAR before output is delivered
BEV-INV-010: CTCHC initialized before first BAR
BEV-INV-011: Each CTCHC link = H(prev turn_hash governing_receipt_id)
BEV-INV-013: Seal hash covers complete chain (first->last link)
BEV-INV-014: CTCHC seal is PQC-signed (ML-DSA-65) before OEP export
MIVP-INV-001: MBR issued before Turn 1 (mandate frozen at session open)
MIVP-INV-003: MAS computed per turn linked to CTCHC link hash
MIVP-INV-007: MBR Seal issued at session close
MIVP-INV-008: MANDATE-BOUND tag on PoGC (admissible path, zero violations+warnings)
CGE-INV-001: CFRs computed at evaluation time (not retroactively)
CGE-INV-002: CAT root hash covers all CFR IDs
CGE-INV-003: fragility_score ? [0.0, 1.0]
CGE-INV-007: CAT PQC-signed before OEP export
TGB-INV-001: TCS embedded in source_state with nanosecond-precision regulatory context
PoGR-INV-001: PoGC issued only from sealed CTCHC session
PoGR-INV-002: PoGC append-only - content_hash immutable after issuance
PoGR-INV-003: PoGC verifiable offline (zero runtime, zero auth)
RTE-INV-001: Package contains both dangerous and admissible paths
RTE-INV-002: Dangerous path terminates in HALT (steps 3+5)
RTE-INV-003: OSG rejects dangerous path independently (fail-closed)
RTE-INV-004: Admissible path PoGC carries MANDATE-BOUND certification
RTE-INV-005: Both paths contain CGE CAT with 5 CFRs
RTE-INV-006: Both paths contain TCS with regulatory_context
RTE-INV-007: Package verifiable offline - zero OMNIX runtime required
RTE-INV-008: CLI verifier exits 0 on PASS, non-zero on FAIL

§13 - Canonicalization Profile & Technical Notes

Canonicalization Profile - ADR-200 §4 + ADR-201

Artifact	Hash	JSON Sep	Sig Payload	Invariant
DR / TAR	SHA-256	compact (',',:')	sign(content_hash.encode())	ATF stack
RCR	SHA3-256	default	compact sig {rcr_id, rcr_hash}	ATF-INV-003
MBR / MAS	SHA3-256	default (excl key)	compact sig {id, hash, issued_at}	MIVP-INV-001/003
MBR Seal	SHA3-256	default	compact sig {seal_id, hash, issued_at}	MIVP-INV-007
CAT	SHA3-256	default	compact sig {cat_id, hash, issued_at}	CGE-INV-007
TCS / Replay	SHA3-256	default	compact sig {id, hash, issued_at}	TGB-INV-001
OSG VR	SHA3-256	default	compact sig {vr_id, hash, issued_at}	OSG-INV-001
Refusal / Outcome	SHA3-256	default	compact sig {receipt_id, hash, issued_at}	RTE-INV
BAR content_hash	SHA3-256	default (6 fields)	default sig {bar_id, hash, governing_receipt_id, created_at}	BEV-INV-002
CTCHC link hash	SHA3-256	default	SHA3({prev, turn, receipt})	BEV-INV-011
CTCHC seal_hash	SHA3-256	7-field payload	default sig {chain_id, seal_hash, session_id}	BEV-INV-013
PoGC	SHA3-256	default (incl alg)	compact sig {pogc_id, hash, issued_at}	PoGR-INV-003

Verifier Scope Limits

OUT OF SCOPE	Governance policy value validation (FX bands, counterparty lists, mandate amounts)
OUT OF SCOPE	External market data in source_state
OUT OF SCOPE	OMNIX runtime, database, or network access
IN SCOPE	Embedded public key produces valid PQC signatures for ALL artefacts (30 sig checks)
IN SCOPE	Hash integrity confirmed for all 14 artifact types
IN SCOPE	Dangerous path HALTED: execution_occurred=False, settlement_released=False
IN SCOPE	Admissible path CLOSED: execution_occurred=True, settlement_released=True
IN SCOPE	CTCHC chain continuity turn-by-turn verification
IN SCOPE	CAT cfr_root_hash covers all 5 CFR IDs
IN SCOPE	CES formula consistency: Tx0.30 + Bx0.30 + Dx0.20 + lx0.20

§14 - Verification Summary & PQC Public Key

Package Summary

Package ID	OMNIX-RTE-001-77B1453F819D4F52
Package Type	OMNIX-RTE-001
Generated At	2026-05-30 10:38:59 UTC
Scenario	Autonomous Treasury Approval
Amount	USD 50,000,000
Settlement Rail	SWIFT MT202 / XRPL RLUSD
PQC Algorithm	ML-DSA-65 (Dilithium-3, FIPS 204)
PQC Key Size PK	0 bytes (Dilithium-3 standard)
PQC Key Size SK	0 bytes
Total Artifacts	14 types . 2 paths . 30 PQC signatures
Invariants	30 demonstrated
Verifier Checks	111 / 111 PASS
ADR Reference	ADR-201 . RFC-ATF-1 -> RFC-ATF-6
FIPS Standard	FIPS 204 (ML-DSA-65 / Dilithium-3)

PQC Public Key (ML-DSA-65) - Embedded for Offline Verification

The following public key is embedded in the package. It is the sole key required to verify all 30 PQC signatures in this package offline. No other key material, network access, or OMNIX runtime is required.

SHA-256 fingerprint	E4D63A7F20A43ECF582C5BD94B2C9D7081E454E9ACB68349763DF8DFF44537B6
SHA3-256 fingerprint	3A9E370FEF603B5CCBCA2663A0A9A8D8054AE7002D032AB490B5670EF5E852FA

Public Key (Base64, first 512 chars):

WcYkLYDLKZLET1Sp5SbwR1+CRbxzhi64r3IhIw5T1RJUSJ0Fjc0HmFrxfST/IyCwjC2IAB00/pJ560WqKhJLzf7Sde80LpvwaDTD
Qvdm/3uo/xwJjYyevpXHLokJv+LcVcgS30uRON17XVcykHHYr8chLqAk0Inutob8JwPUC4mjf4Z3rjq9nvIJKl+sKP820m0GraU+
JGLxtE8tZa1w0kJccXvC005w+2x8zvQZqvWYEp01PGkSrcIuTMAw5fKZzUncFp5X9dAjV42vtg8+pUA5fKcvGwh/jTAMAbDMvPF+
wk7kToyH+WGv9V08MVxf0bdjv3XPoNZtSapV0k80B9aG2xxxii+qjGwQ7YJ9a5yZZiHwpUaAcZ/mw+Qs9eV0pVw202UgwIYyt/MmK
PG+p4Pm0xtGs6sd7iFii+cpqTzHBP271y0Ly91tS0vCs25MljAiK4wC1myF8VsdwRrrqbVM+Z56Q4v9EkABH3ZNxzY/cQxfdsLGb
9JtN0mj1GMap

Governance Verdict

PATH DANGEROUS: HALT PROVEN	PATH ADMISSIBLE: SETTLEMENT RELEASED
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OMNIX-RTE-001 . 111 / 111 VERIFICATIONS PASS . RFC-ATF-1 -> RFC-ATF-6